

Converters Educational Platform

BUCHAREST
2026

Project Content

- Digital to Analog converter – DAC0808
- SAR Analog to Digital converter – ADC0804
- $\Delta\Sigma$ Analog to Digital converter – ADS1110
- Voltage to Frequency converter – LM331

All mounted on PCB, equipped with its own power supply.

Micro DER Lab



Power Supply



D/A Converter



V/F Converter



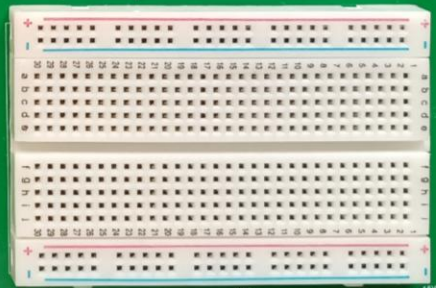
Delta-Sigma ADC



SAR ADC



Experiment



Converters Sections



Switch + LED

Fuse

$\pm 24\text{VDC}$

-15VD

+15VDC

10VDC

5VDC

Converters Sections

DAC0808



LM 741 OP Amp

Arduino Nano

Test points + ext. input jumper (center)

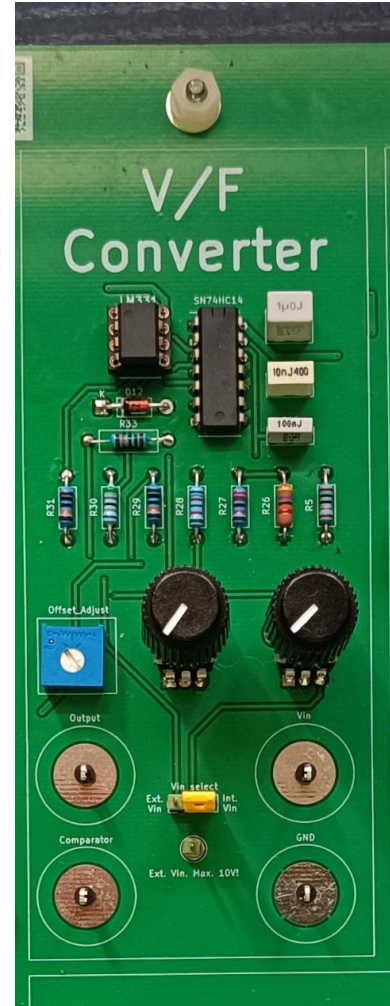
Input status LEDs

External input pins

3 × 4 Keypad

Converters Sections

LM331



Schmitt-Trigger

Capacitors &
Resistors

Offset trimmer (left), Gain pot. (center), Vin (right)

Input & output test points

Center: jumper for internal/external Vin + pin for external Vin

Converters Sections

Diff./single-ended input (left), trimmer/DAC0808 Vin-
jumper (center), trimmer, Vin+ pot. (right)

Vin-, Vin+, SDA, SCL test points

Center: internal/external Vin+ jumper & ext. input pin



ADS1110

Arduino nano + display

Converters Sections

ADC0804 (center)

Output status LEDs

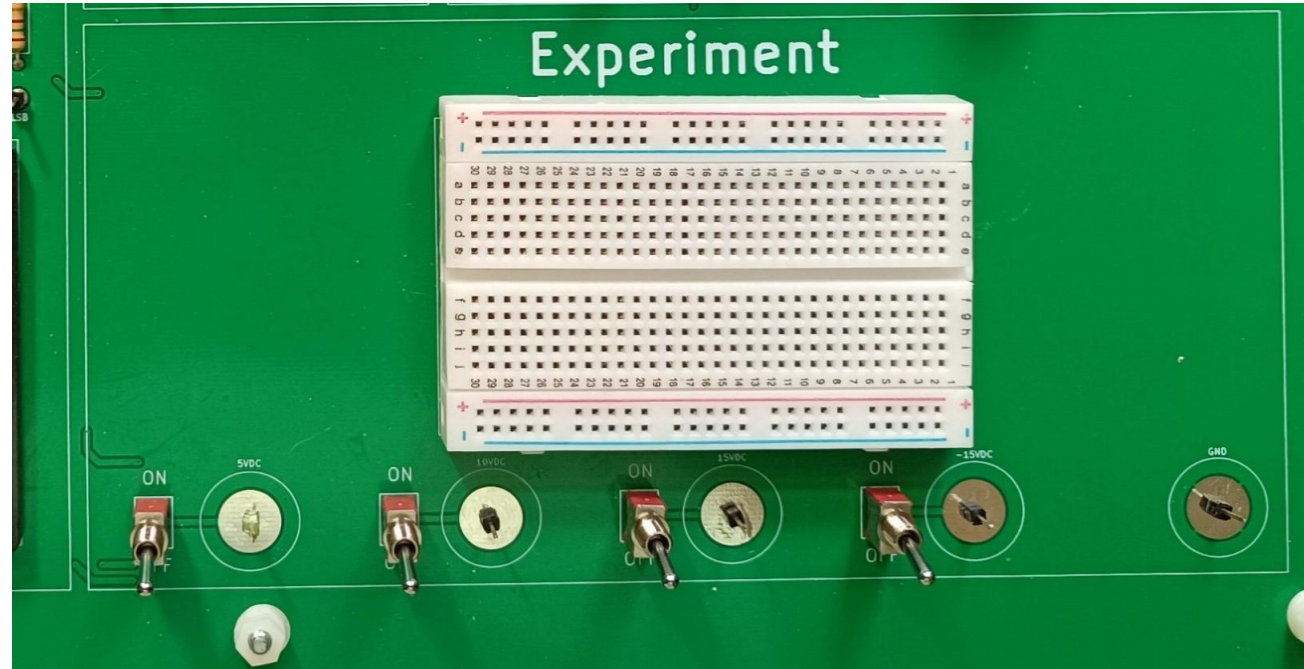
Start(left), Vin+ (center), int/ext. jumper & ext. Vin+ pin
(green), GND/DAC0808 diff. input jumper (yellow)

Test points: Vin+, Vin-, Interrupt



8 ch. Darlington
bridge

Converters Sections



5VDC

10VDC

15VDC

-15VDC

GND

About the Converters

- **DAC0808:**

- 150ns settling time
- Full scale output: 10V
- 8-bit resolution $\rightarrow$ LSB = 39mV
- Accuracy: $< \pm 0.19\%$ FS $\rightarrow \pm 19.5\text{mV}$ max. deviation

- **ADC0804:**

- 100 μ s conversion time $\rightarrow$ sampling freq.: 10kSPS
- Full scale input: 5V
- 8 bit resolution $\rightarrow$ LSB = 19.53mV
- Total error: ± 1 LSB
- Single-ended or differential input

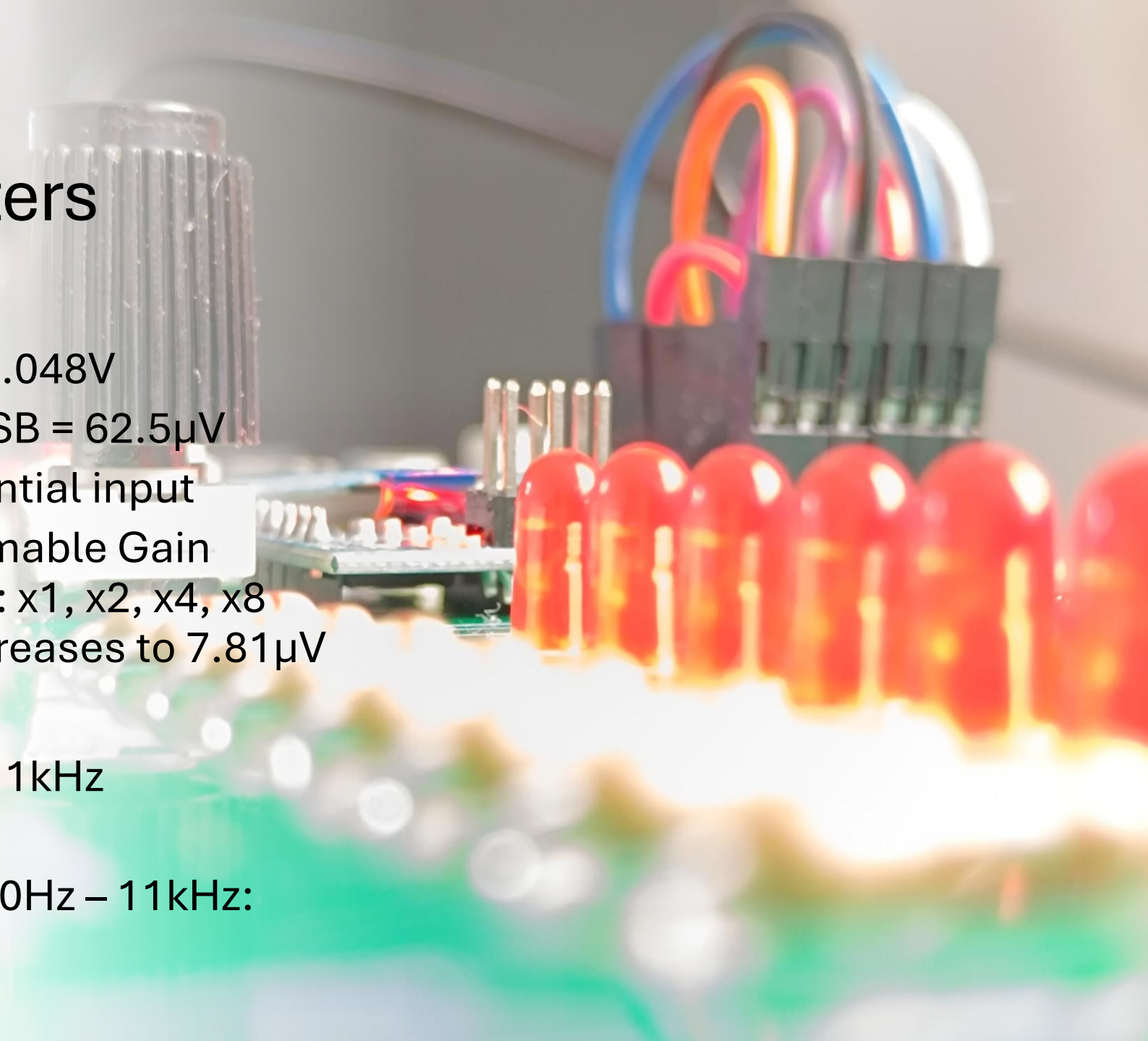
About the Converters

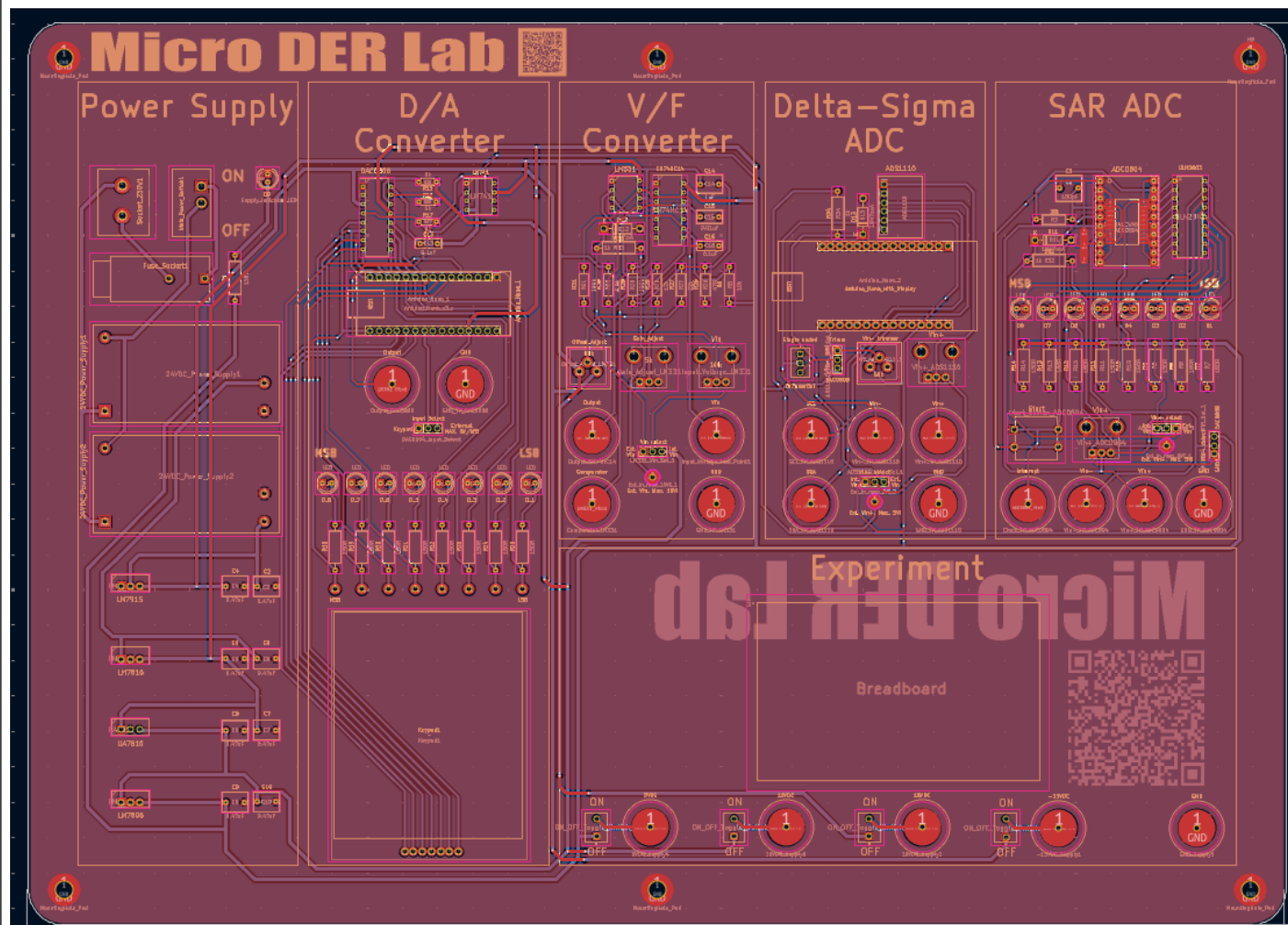
- **ADS1110:**

- input voltage range: $\pm 2.048\text{V}$
- Resolution: 16bits $\rightarrow$ $\text{LSB} = 62.5\mu\text{V}$
- Single-ended or differential input
- internal PGA (Programmable Gain Amplifier) with gains of: x1, x2, x4, x8
For x8 $\rightarrow$ resolution increases to $7.81\mu\text{V}$

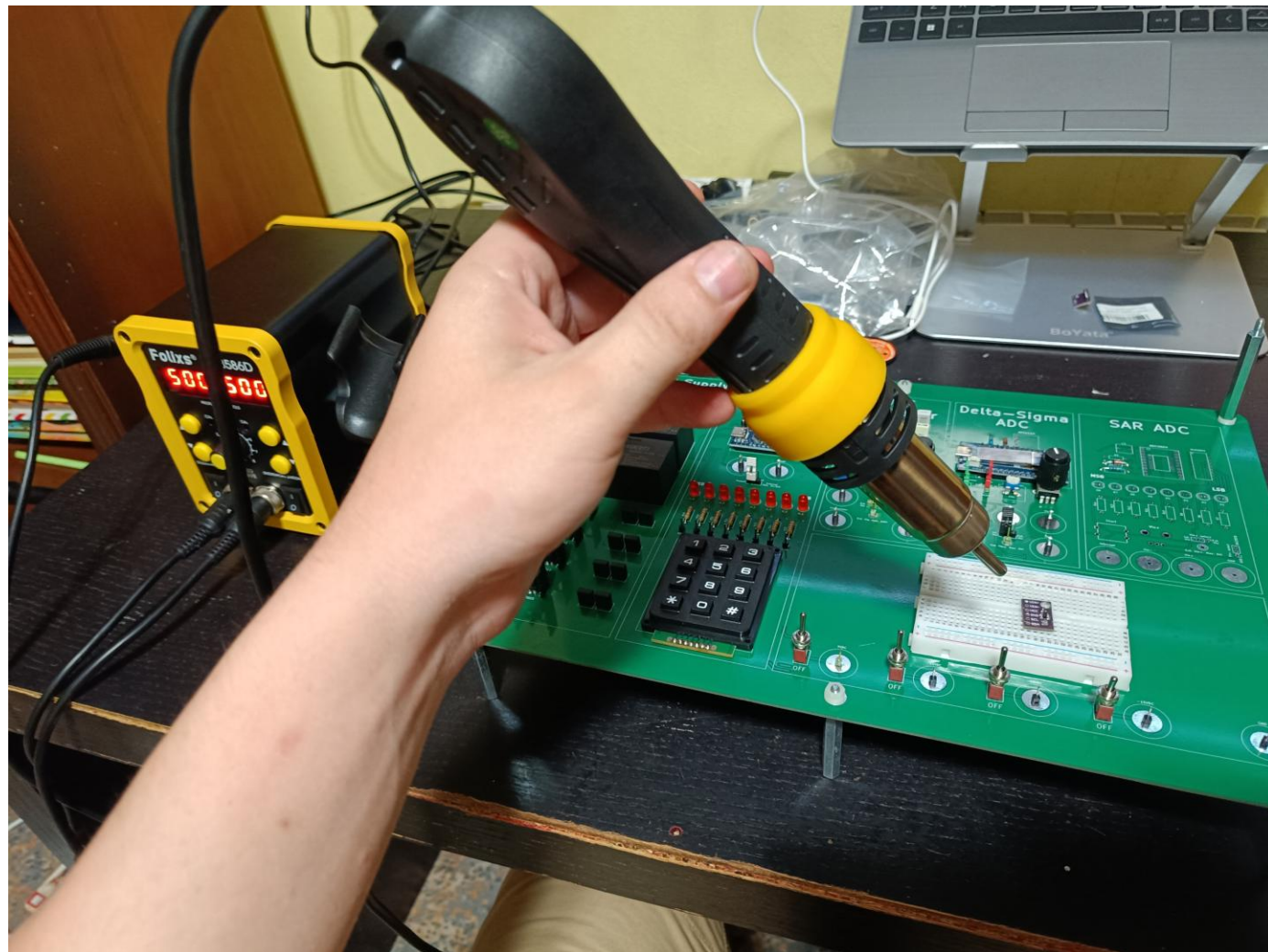
- **LM331:**

- Configuration: 10Hz – 11kHz
- Linearity: $\pm 0.03\%$ of FS
- Dynamic range for FS 10Hz – 11kHz:
81.57dB





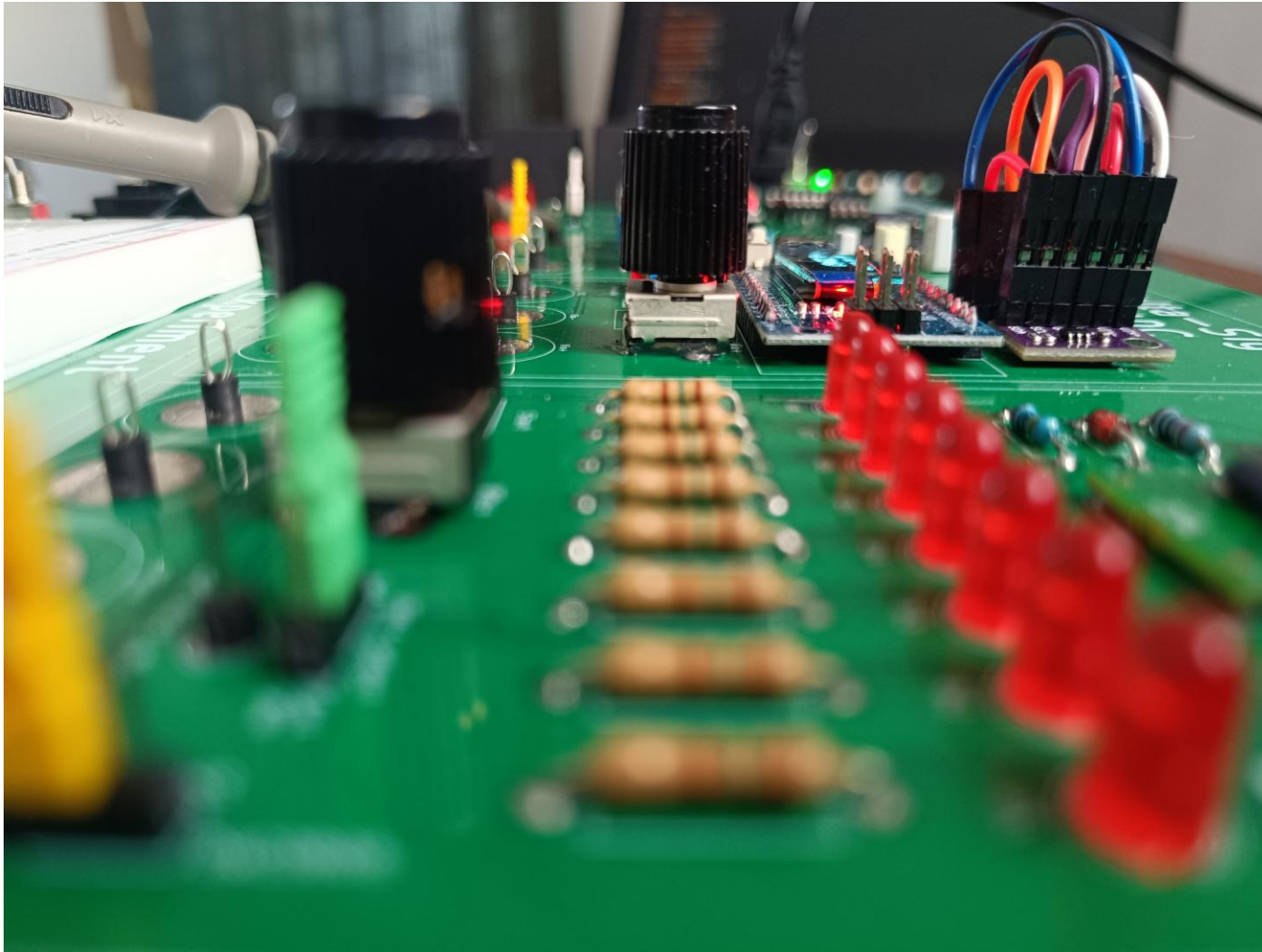
Photos



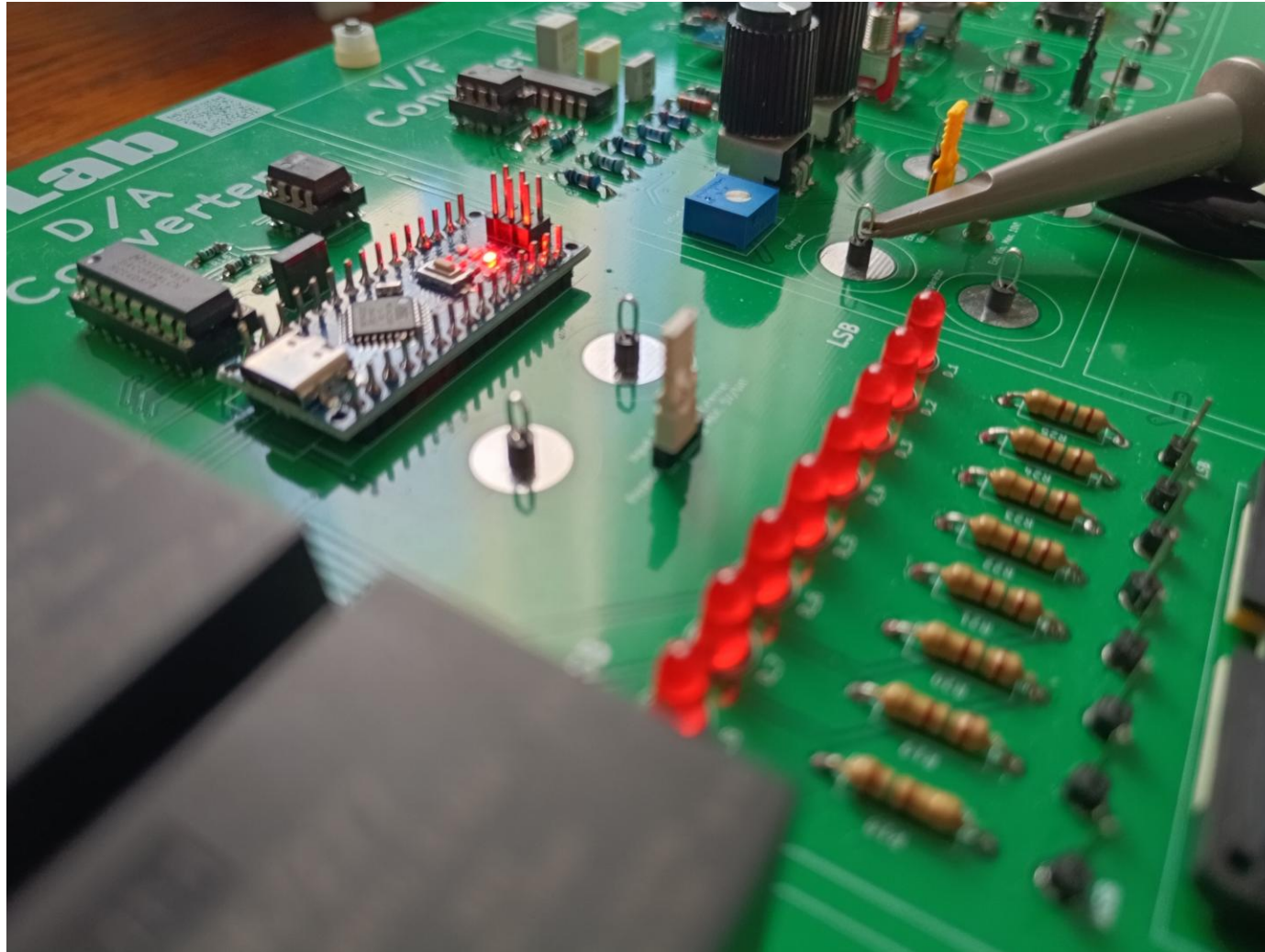
Photos



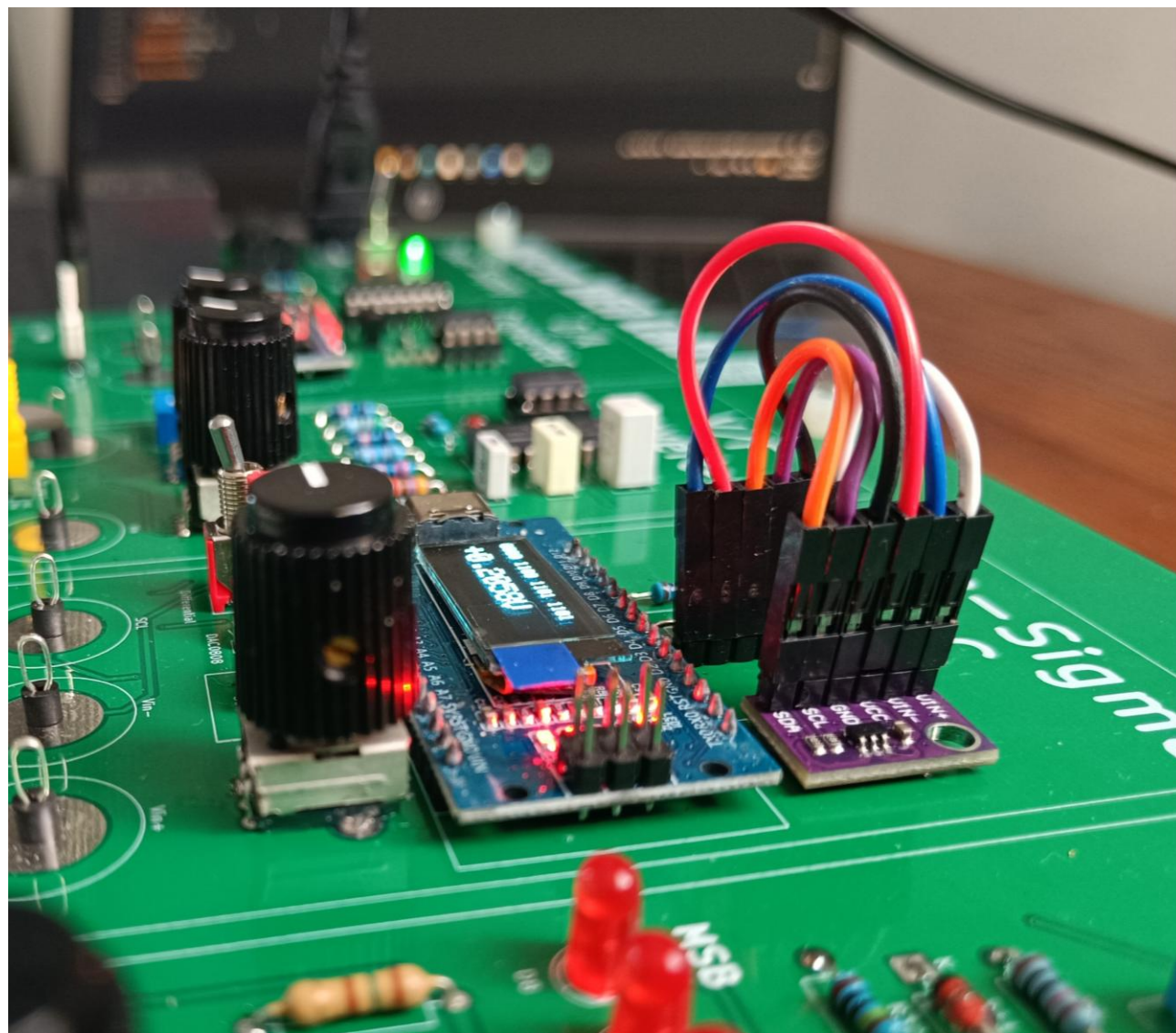
Photos



Photos



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